

Minor Project Report
On
Recommendation System using Association Rule
Mining based on Contrastive Learning

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Declaration

We solemnly declare that the interim project report 'Recommendation System using Association Rule Mining-based On Contrastive Learning' is based on our own work carried out during the course of our study in the Department of Computer Science & Engineering under the supervision of Dr. Somaraju Suvvari, Assistant Professor, Department of Computer Science and Engineering, National Institute of Technology Patna. We assert the statements made and conclusions drawn are an outcome of our work. To the best of our knowledge, we certify that the work contained in the report is original and has been done by us under the general supervision of our supervisor. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and given their details in the references.

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Chapter 1

Introduction

1.1 Background

In today's digital world, people are surrounded by huge amount of information. Whether shopping online, streaming movies, reading articles, or listening to music, finding what truly matches personal tastes can feel like searching for a needle in a haystack. Recommendation systems have become essential tools to solve this problem by suggesting items tailored to individual preferences. For example, Amazon recommends products based on past purchases, Netflix suggests shows that align with viewing history, Spotify curate's playlists reflecting listening habits, and news platforms highlight articles that match user interests. These systems help users navigate vast digital content, making their experiences more enjoyable and efficient.

Recommendation systems play an important role in enhancing the user experience across industries. They filter through extensive data to discover patterns in user-item interactions, ensuring suggestions are both relevant and engaging. Using advanced techniques such as machine learning (ML) and deep learning (DL), these systems achieve high levels of personalization. This report explores the mechanics of recommendation systems, focusing on a novel hybrid approach that integrates contrastive learning with positive and negative association rules to improve recommendation accuracy. The objective is to develop a system that provides precise and diverse suggestions that address the problem of information overload while enhancing user satisfaction. Our contribution lies in combining contrastive learning with association rule mining to create enhanced user-item representations, leading to more effective recommendations.

1.2 Recommendation System Approaches

Recommendation systems rely on machine learning and deep learning to understand what users like and predict what they might want next.

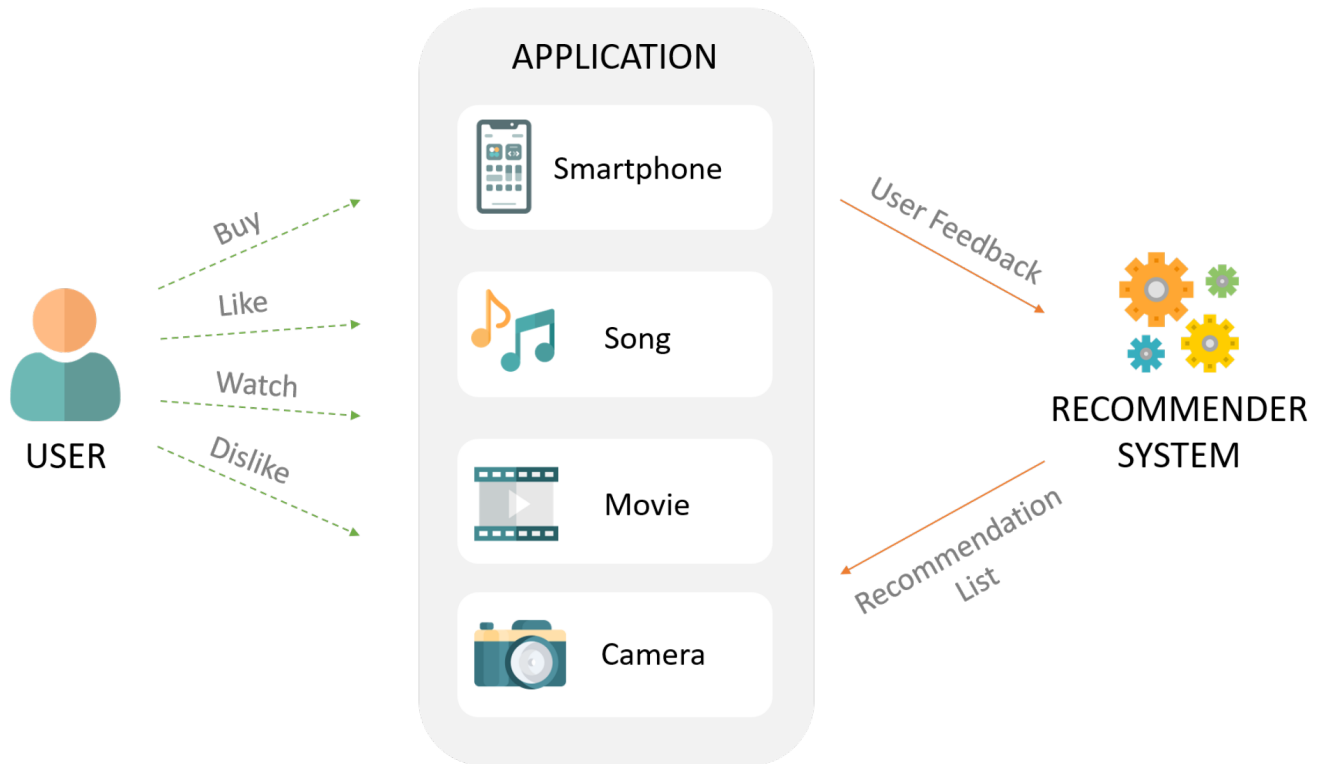


Figure 1.1: Overview of a recommendation system.

One approach, Neural Collaborative Filtering, improves on traditional methods by using deep neural networks to capture complex user behaviors. Unlike older techniques that assume simple relationships, it uses layers of transformations to learn that, for instance, a user buying running shoes might also want fitness trackers. However, it demands large datasets and significant computing power, and it can struggle with sparse data, where user interactions are limited, sometimes overfitting to overly specific patterns.

Another approach, Graph Neural Networks (GNNs), treats users and items as nodes in a graph, with interactions as connecting edges. GNNs learn by gathering information from nearby nodes, capturing complicated relationships, such as recommending movies to users who share similar viewing habits. This method shines in sparse data scenarios, but is computationally heavy and complex to scale for platforms with millions of users or items.

1.3 Contrastive Learning

Contrastive learning is an emerging technique that enhances recommendation systems by improving the quality of user and item representations. It works by learning to distinguish between positive (relevant) and negative (irrelevant) pairs of data points. In the context of recommendations, contrastive learning optimizes embeddings such that similar users or items are closer in the representation space, while dissimilar ones are farther apart.

This approach is particularly effective in scenarios with sparse data, as it leverages both positive interactions (e.g. purchases) and negative feedback (e.g., ignored items) to refine predictions.

By incorporating contrastive learning, recommendation systems can better capture user preferences. The technique also mitigates issues like popularity bias, where systems over-recommend popular items, by focusing on personalized patterns. Our project utilizes contrastive learning to enhance the robustness of item embeddings, ensuring more accurate and diverse recommendations.

1.4 Association Rule Mining

Association Rule Mining identifies interesting patterns and correlations between items in large datasets. Within recommendation systems, it is particularly useful for discovering implicit relationships among items based on user behavior.

Positive association rules identify patterns such as 'Users who buy bread also buy butter', indicating items that are often purchased or interacted with together. These rules improve the system's ability to suggest complementary or similar items, thus improving user satisfaction and cross-selling opportunities.

In contrast, negative association rules recognize exclusions, such as "Users who buy bread rarely buy detergent." These insights allow the system to avoid suggesting irrelevant or potentially conflicting items, enhancing the relevance and efficiency of recommendations.

By integrating both types of association rules into recommendation systems, we can enhance the quality of suggestions, support content diversity, and fine-tune the personalization process to avoid redundancy or user dissatisfaction [2].

1.5 Problem Statement and Objective

Recommendation systems face significant challenges in accurately predicting user preferences from historical interaction data, particularly in domains with large item catalogs and sparse user interactions, such as retail or e-commerce. Traditional next-item prediction models, which often rely on collaborative filtering or mainly positive association rules, struggle to capture the full complexity of user behavior. These models typically emphasize items that frequently co-occur, overlooking negative associations (items rarely purchased together) that provide essential context for understanding user preferences. This limitation hampers the ability to model complicated sequential patterns and item relationships, resulting in recommendations that lack precision and personalization.

The objective of this project is to develop an advanced recommendation system that

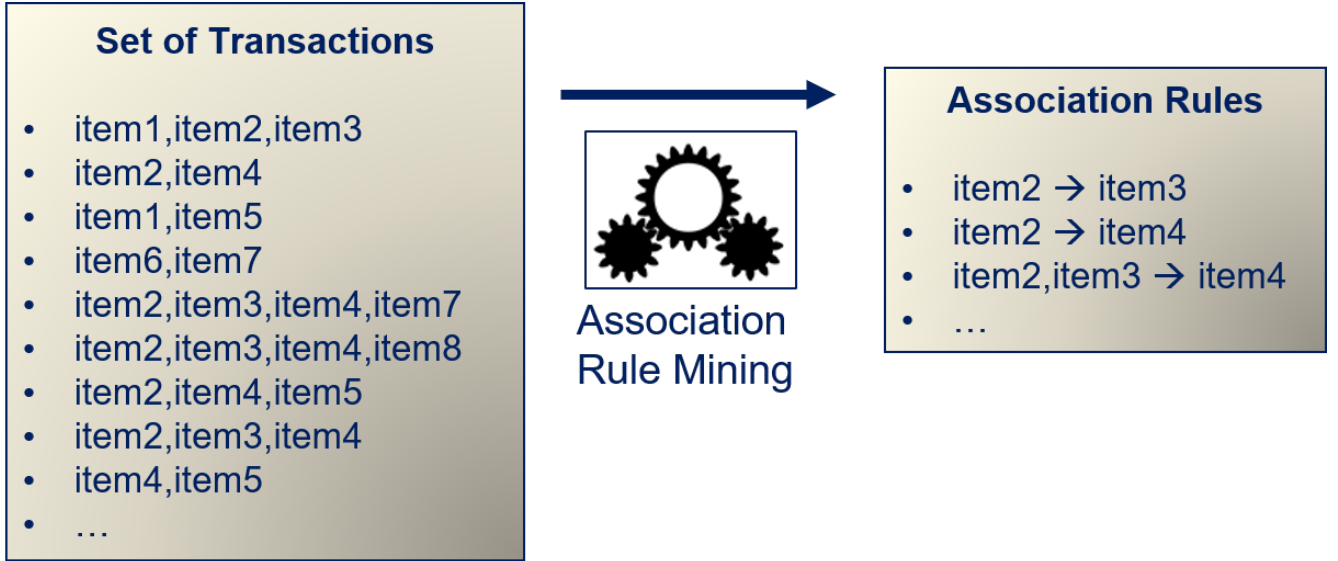


Figure 1.2: Association rule mining for item relationships.

employs positive and negative association rules within a contrastive learning framework to generate high-quality embeddings of items. By extracting co-occurring and mutually exclusive item relationships from user interaction data, the system aims to create comprehensive embeddings that capture a rich representation of item interactions. These embeddings are designed to be modular, enabling their potential integration into various sequential recommendation architectures, such as transformer-based models like SASRec, recurrent neural networks, or other deep learning frameworks, to effectively model sequential user behavior. The performance of the system is evaluated using standard metrics such as Recall@k and NDCG@k to ensure the delivery of accurate and personalized recommendations. The ultimate goal is to create a scalable recommendation framework that aims to improve traditional next-item prediction models by incorporating both positive and negative association rules through contrastive learning, thereby enhancing recommendation accuracy and user experience in various recommendation scenarios.

1.6 Contribution and Achievement

This project significantly advances the field of recommendation systems by introducing a novel framework that integrates both positive and negative association rules within a contrastive learning approach to generate high-quality item embeddings. A key contribution is the development of a contrastive learning pipeline that leverages association rules extracted from user interaction data to create embeddings that capture both co-occurring and mutually exclusive item relationships. Unlike traditional methods that focus solely on positive co-occurrence patterns, this system incorporates negative rules to model item dissimilarities, resulting in richer and more enhanced embeddings. This is achieved through the implementation of the ItemEncoder and a ContrastiveLoss function, which

optimizes the embedding space to reflect complex item interactions.

Another major contribution is the modular design of the embedding generation process, which enables the embeddings to be utilized across a wide range of advanced machine learning and deep learning models. The embeddings, stored as a standalone NumPy array, are not restricted to sequential recommendation architectures but can be integrated into various models, including transformer-based architectures (e.g., SASRec), graph neural networks, recurrent neural networks, or other supervised and unsupervised ML/DL frameworks. To demonstrate this versatility, the project applies the embeddings to a transformer-based SASRec model, showcasing their effectiveness in sequential recommendation tasks, while their format ensures compatibility with other advanced architectures for diverse recommendation scenarios.

The achievements of this project are highlighted by the successful implementation and evaluation of the recommendation system. The system was trained and validated on a retail dataset, achieving consistent improvements in recommendation quality, as evidenced by the Normalized Discounted Cumulative Gain (NDCG@10) metric, which improved from 0.0630 in the first epoch to 0.0755 by the tenth epoch. The evaluation process, employing metrics such as Recall@k and NDCG@k (for $k=5$ and 10), confirms the system’s ability to deliver accurate and personalized recommendations. Furthermore, the incorporation of early stopping and learning rate scheduling (ReduceLROnPlateau) ensured robust training, optimizing performance and preventing overfitting.

Overall, this project contributes a scalable and flexible recommendation framework that enhances the state-of-the-art by combining association rule mining with contrastive learning. The resulting embeddings and their adaptability to various advanced ML/DL models pave the way for improved recommendation accuracy and user experience in retail, e-commerce, and other domains, setting a foundation for future research into hybrid recommendation systems.

Chapter 2

Literature Review

The development of recommendation systems has been a vibrant area of research, with significant advancements driven by machine learning and deep learning techniques. This literature review synthesizes key works in neural collaborative filtering, graph-based methods, contrastive learning, and association rule mining, highlighting their contributions, limitations, and relevance to the current project, which integrates positive and negative association rules with contrastive learning for recommendation systems.

Neural Collaborative Filtering and Deep Learning Approaches

He et al. (2017) introduced Neural Collaborative Filtering (NCF), a seminal work that replaced the linear assumptions of traditional matrix factorization with a deep learning architecture [4]. By modeling user-item interactions through neural networks, NCF captured complex, non-linear patterns, achieving higher recommendation accuracy than conventional methods. However, its reliance on dense interaction data posed challenges in sparse data scenarios, and the computational complexity increased the risk of overfitting, particularly in resource-constrained environments. This work laid the foundation for deep learning in recommendations but highlighted the need for methods that handle sparsity effectively, a challenge addressed in part by the current project’s use of association rules.

Graph-Based Recommendation Systems

Wang et al. (2019) advanced the field with Neural Graph Collaborative Filtering (NGCF), leveraging graph neural networks (GNNs) to model high-order connectivity in user-item interaction graphs [5]. NGCF exploited the relational structure of data to generate richer embeddings, improving recommendation quality over traditional collaborative filtering. Despite its strengths, the computational demands of GNNs and the complexity of graph construction limited its scalability, particularly for large-scale datasets with dynamic interactions. While NGCF’s focus on structural relationships is valuable, it does not explicitly incorporate explicit item associations, an area where the current project’s use of positive and negative association rules offers complementary insights.

Contrastive Learning in Representation Learning

Contrastive learning has emerged as a powerful self-supervised approach for representation learning, with significant implications for recommendation systems. Oord et al. (2018) introduced Contrastive Predictive Coding (CPC), a method that predicts future data segments from past ones without requiring labeled data [6]. Although designed for general representation learning, CPC’s principles inspired subsequent recommendation research by demonstrating the potential of self-supervised learning to generalize across tasks. However, its application to recommendations required adaptation to capture domain-specific patterns. Xie et al. (2020) addressed this by applying contrastive learning directly to recommendation systems, forming positive and negative user-item interaction pairs to enhance robustness in cold-start and sparse data scenarios [7]. Their approach improved recommendation quality, but its effectiveness depended heavily on pair construction and data augmentation strategies, leading to variable performance. These works underscore the potential of contrastive learning, which the current project extends by using both positive and negative association rules to guide the contrastive learning process, ensuring more nuanced item embeddings.

Association Rule Mining in Recommendations

Association rule mining has been a cornerstone of recommendation systems due to its simplicity and interpretability. Aggarwal et al. (2016) explored the use of both positive and negative association patterns to generate recommendations, emphasizing their ability to increase diversity and capture item relationships [8]. Positive rules identified frequently co-occurring items, while negative rules highlighted mutually exclusive pairs, offering a more comprehensive view of item interactions. However, the approach struggled to scale with highly individualized user preferences and often produced generic rules that lacked deep personalization. Additionally, traditional association rule mining methods were not seamlessly integrated with modern deep learning frameworks, limiting their ability to model complex sequential patterns. The current project addresses these limitations by combining association rule mining with contrastive learning, using both positive and negative rules to initialize and refine item embeddings within a deep learning pipeline.

Relevance to the Current Project

The reviewed literature provides a strong foundation for the current project, which builds on the strengths of contrastive learning and association rule mining while addressing their limitations. By integrating positive and negative association rules into a contrastive learning framework, the project creates embeddings that capture both co-occurring and mutually exclusive item relationships, enhancing recommendation accuracy. Unlike Aggarwal et al. (2016), which relied solely on rule-based recommendations, this project combines rules with deep learning to model complex user behaviors [8]. Compared to Xie

et al. (2020), the use of association rules provides a more structured approach to pair construction, reducing variability in contrastive learning performance [7]. Furthermore, the modular embeddings are designed to be compatible with various advanced models, extending beyond the sequential focus of prior works like Wang et al. (2019) [5]. By addressing the gaps in scalability, personalization, and integration, this project contributes a novel and versatile recommendation framework.

Chapter 3

Methodology

To build our recommendation system, we first collected user-item interaction data, capturing items users have liked, bought, or interacted with. We cleaned this data to remove extraneous or missing information, ensuring the model could learn effectively.

We then applied association rule mining to identify patterns in item interactions, such as items frequently purchased together or avoided. This process generated positive rules (indicating co-purchases) and negative rules (indicating exclusions), which informed the subsequent learning phase .

Next, we employed contrastive learning to create robust item embeddings. By comparing positive and negative item pairs derived from the association rules, the model learned to position similar items closer and dissimilar items farther apart in the embedding space. These embeddings captured nuanced item relationships critical for accurate recommendations .

Finally, we utilized a Transformer-based SASRec model to process user interaction sequences and predict the next item. The model leveraged the learned embeddings to understand temporal patterns in user behavior . We evaluated its performance by checking if the recommended items matched users' actual preferences, confirming the system's ability to deliver personalized suggestions through the integration of association rule mining, contrastive learning, and sequential modeling.

3.1 Datasets

3.1.1 Dunnhumby

Source: Transaction records from a retail chain (grocery-focused).

(<https://www.dunnhumby.com/source-files/>)

Structure: CSV format with three columns: `CUSTOMER_ID` (unique, 1 to 5632), `ORDER_NUMBER` (sequential per customer), `MATERIAL_NUMBER` (product ID, 2 to 8332).

Scale: Large dataset, over 7.7 million characters, millions of transaction records.

Characteristics: Transaction-level data with multiple orders per customer and frequent product purchases, ideal for segmentation and recommendation systems.

3.1.2 ValuedShoppers

Source: Acquire Valued Shoppers Challenge

(<https://www.kaggle.com/c/acquire-valued-shoppers-challenge>).

Structure: CSV with three columns: `CUSTOMER_ID` (0 to 649), `ORDER_NUMBER` (sequential, e.g., 0 to 279), `MATERIAL_NUMBER` (product ID, 1 to 10823).

Scale: Large, over 7.6 million characters, numerous transaction records.

Characteristics: Captures multiple customer orders with frequent products (e.g., 80, 183), suitable for market basket analysis and customer segmentation.

3.1.3 Ta-Feng

Source: Ta Feng Grocery Dataset.

(<https://www.kaggle.com/datasets/chiranjivdas09/ta-feng-grocery-dataset>)

Structure: CSV with three columns: `CUSTOMER_ID` (0 to 9628), `ORDER_NUMBER` (sequential, up to 279), `MATERIAL_NUMBER` (product ID, 0 to 10800).

Scale: Millions of records, at least 9629 customers, 10801 products, over 7.6 million characters.

Characteristics: Sparse transactional data with repetitive purchases (e.g., 70, 203), supports analysis of purchase patterns and recommendation systems.

3.2 Data Preparation

The data preparation phase involves loading the data set into a structured format, ensuring error-free data integrity. Unique elements are identified to construct a vocabulary, assigning numerical indices for embedding and model training. User transactions are organized into sequences to capture purchasing behavior, potentially limiting to a subset of users (e.g., the first 200) for computational efficiency during testing.

3.3 Association Rule Mining

Association rule mining begins by calculating support and confidence to establish frequent item co-occurrences, caching these for efficient rule generation. The phi-coefficient measures item pair correlations, identifying positive and negative associations. Rules are generated for both strongly co-occurring items and those rarely appearing together, with redundant or weaker symmetric rules filtered out to retain only the most informative ones based on

confidence.

3.4 Learning Item Embeddings via Contrastive Learning

Item embeddings are learned using a neural network encoder that maps items to dense vectors (e.g., 16 dimensions). Training pairs are created from positive and negative associatirules andes, transformed into numerical indices. A contrastive loss function pulls positive pairs closer and pushes negative pairs apart in embedding space, using cosine similarity. The resulting embeddings form the basis for subsequent recommendation tasks.

3.5 SASRec-Based Sequential Recommendation Model

The SASRec model[1] processes time-sorted user interaction sequences, using the last item for testing and the rest for training, excluding users with short histories. A dataset structure with padding and batching supports efficient training. The Transformer-based SASRec model, initialized with learned item embeddings, uses self-attention and a final prediction layer. Trained with cross-entropy loss and the Adam optimizer, its performance is evaluated using Recall@k and NDCG@k (e.g., k=5, 10) to assess recommendation accuracy and ranking quality.

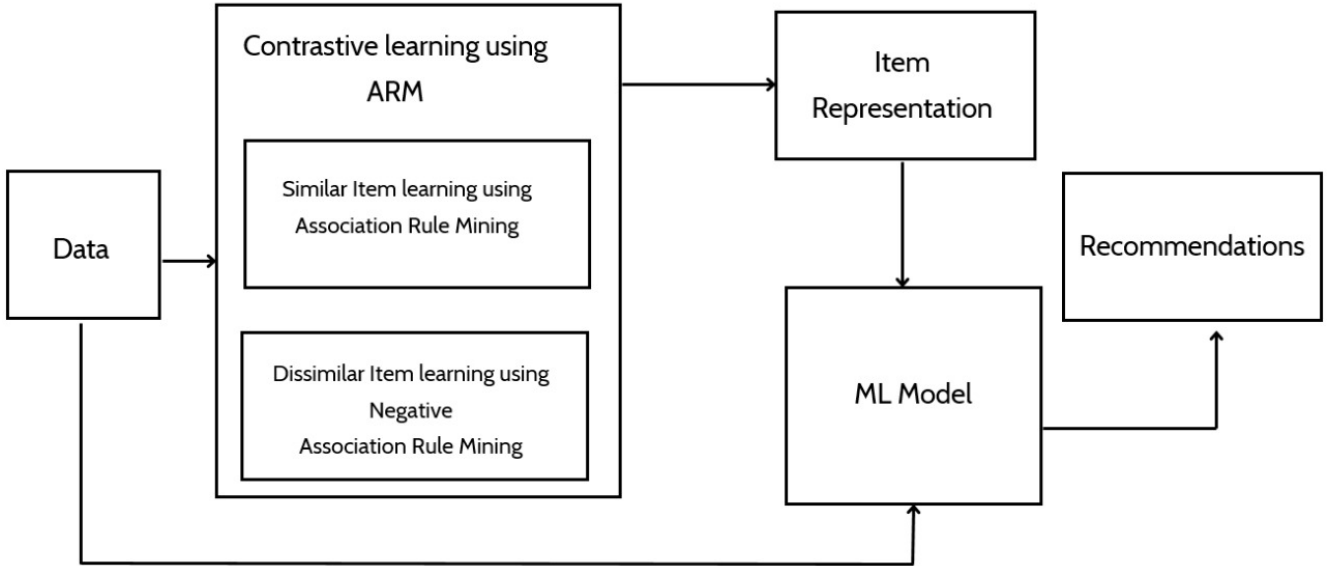


Figure 3.1: Workflow of the proposed recommendation system.

Chapter 4

Results and Analysis

The proposed recommendation system, combining association rule mining, contrastive learning, and SASRec-based sequential modeling, was evaluated on the TaFeng, Dunnhumby, and ValuedShopper datasets. Below are the key results:

4.1 SASRec Performance

Trained for 10 epochs with a batch size of 64 and learning rate of 0.001.

Table 4.1: Performance of SASRec Model on Datasets

Dataset	Recall@5	Recall@10	NDCG@5	NDCG@10
Ta-Feng	0.0624	0.0835	0.0510	0.0577
Dunnhumby	0.1081	0.1357	0.0817	0.0906
ValuedShopper	0.0600	0.0862	0.0405	0.0488

4.1.1 Quantitative Comparison

Table 4.2: Comparison of methods across datasets (percentages)

Method	TaFeng				Dunnhumby				ValuedShopper			
	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10	R@5	R@10	N@5	N@10
POP	6.47	7.60	8.94	8.66	8.61	10.60	9.45	9.32	3.67	5.06	7.32	6.55
Per-POP	7.40	11.15	10.10	10.50	14.61	19.62	16.33	17.34	15.00	20.97	25.98	24.96
BPMC	6.17	7.30	6.59	6.46	10.28	7.88	8.78	9.41	3.56	5.54	5.19	5.50
CLEAM	6.71	8.05	9.39	9.56	8.97	10.03	10.14	10.53	3.32	4.84	6.32	6.50
SASRec	6.24	8.35	5.10	5.77	10.81	13.57	8.17	9.06	6.00	8.62	4.05	4.88

4.2 Analysis

- The SASRec model effectively captured sequential patterns in user purchase histories, leveraging high-quality embeddings from contrastive learning.

- The high number of positive rules suggests strong co-purchase patterns in the dataset, which contrastive learning encoded into robust embeddings.
- Negative rules, though fewer, provided critical contrastive signals to differentiate dissimilar items.
- Low recall and NDCG scores indicate challenges with sparse data and the large item set, limiting prediction accuracy.
- The accuracy of a recommendation system can be improved by increasing the number of epochs. We used 10 epochs due to hardware constraints.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This work presents a novel hybrid recommendation system that integrates association rule mining, contrastive learning, and sequential modeling with the SASRec model. By combining positive and negative association rules with contrastive learning, the system generates robust item embeddings that capture both co-occurrence and exclusion patterns. The SASRec model effectively leverages these embeddings to deliver personalized recommendations. Evaluated on the TaFeng, Dunnhumby, and ValuedShopper datasets, the system demonstrates reasonable performance, with Recall@10 scores ranging from 0.0835 to 0.1357 and NDCG@10 scores from 0.0488 to 0.0906. These results highlight the system’s ability to handle diverse retail transaction data.

The integration of negative association rules is a key differentiator, enabling the system to avoid irrelevant suggestions and enhance recommendation diversity. Contrastive learning further strengthens the model by improving embedding quality, particularly in sparse data scenarios. Despite limitations, such as low recall due to large item sets and hardware constraints limiting training to 10 epochs, the system shows significant promise for real-world applications. This work lays a strong foundation for advancing recommendation systems, offering a balanced approach that combines interpretability, personalization, and robustness.

5.2 Future Work

The current project effectively combines positive and negative association rules with contrastive learning to generate versatile item embeddings for recommendations, as demonstrated with SASRec. Future work could explore hyperparameter optimization to fine-tune key parameters, such as learning rate, embedding dimensions, or contrastive loss margins, to maximize recommendation accuracy and training stability. Integrating

these embeddings into other advanced models, such as graph neural networks or recurrent neural networks, could leverage diverse architectures to improve performance in sparse data scenarios. Enhancing evaluation by comparing against baseline models and testing on diverse datasets would quantify the framework’s effectiveness and generalizability. Improving computational efficiency through model pruning or online learning could enable real-time recommendations and scalability for larger datasets.

Chapter 6

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